DRUGS · SPECIAL ENT AGENTS

Intrathecal fluorescein

Intrathecal fluorescein stains cerebrospinal fluid bright yellow-green, so that a leak from the skull base or ear can be seen during endoscopic repair. It is widely used and helpful, but it is not licensed for intrathecal use, serious neurological complications have occurred at higher doses, and the patient must be consented specifically. Dilute, low-dose technique makes complications rare.

1 CHAPTER ONE At a glance

The agent on one page.

1.1 Summary

INTRATHECAL FLUORESCEIN

	SUMMARY
Agent	Sodium fluorescein (preservative-free 10% or 5% solution)
How it works	A fluorescent dye that mixes with CSF and appears at the site of a leak; a blue-light filter on the endoscope increases sensitivity
ENT uses	Localising CSF leaks at endoscopic repair (spontaneous, traumatic, iatrogenic, encephaloceles); confirming a watertight repair
Dose	Commonly 0.1 ml of 10% fluorescein (10 mg) diluted in 10 ml of the patient's own CSF, injected slowly over about 10 minutes through a lumbar drain or needle, about 30 to 60 minutes before the endoscopic search
Consent	Off-label use; explain the rare risks
Avoid	Higher concentrations or doses; known fluorescein allergy

2 CHAPTER TWO Using it in ENT

Practice.

2.1 Technique

After anaesthesia, insert a lumbar drain (also useful post-operatively in some patients), withdraw CSF, dilute the fluorescein in it, and inject slowly. Head-down positioning and a Valsalva manoeuvre help the dye reach the leak. Fluorescein may take time to appear; a negative search does not exclude a leak. Other localisation tools are CT cisternography, high-resolution CT, MRI cisternography and beta-2 transferrin or beta-trace protein testing of nasal fluid.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Seizures, lower limb weakness and numbness, opisthotonus, cranial nerve deficits, and very rarely death, mostly with higher doses or concentrations; headache and nausea; transient yellow discoloration of skin and urine with systemic absorption.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

> **WHAT THE EXAMINER IS LOOKING FOR**
>
> - The dose and dilution.

- Its off-label status and complications.
- Other ways to localise a CSF leak.

4.2 Common pitfalls

COMMON PITFALLS

- Injecting undiluted or high-dose fluorescein quickly.

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